Cloud Computing

Prof. Davide Bilò

Course information

The lecturer

- Full name: Davide Bilò
- Email: davide.bilo@univaq.it
- Room: 220, “Alan Turing” building
- Office hours: Thursday, 14.30-16.30
- Online meeting: on Teams by appointment

Course info

- Semester: second
- ETCS: 6 Credit points
- 24 lectures
  - Monday, 11.30-13.30 in room A.1.1
  - Thursday, 11.30-13.30 in room C1.16
- Team code: r373i3t

Contents

1. **Thinking in parallel**

   (How to design algorithms for multi-processor systems)

   - models
   - the PRAM model: basic techniques, lists, searching, merge, and more
   - MPC model and basic results

2. **Data stream processing**

   (Algorithms that compute useful information on big data)

   - models
   - basic streaming and semi streaming algorithms

Course objectives and summative assessment

- COURSE OBJECTIVES
  - provide the **skills** to design **algorithms** and **data structures** for **big data problems** using **cloud computing technologies**
    - learn how to design parallel algorithms with good performance on big data
    - learn how to compute useful information for streams of data
- SUMMATIVE ASSESSMENT
  - **Oral examination**
    - Duration: ≈45 minutes
    - Criteria of evaluation
      - the level of knowledge of the course topics
      - the property of use of the scientific language
      - the clarity and completeness of explanations

Suggested references

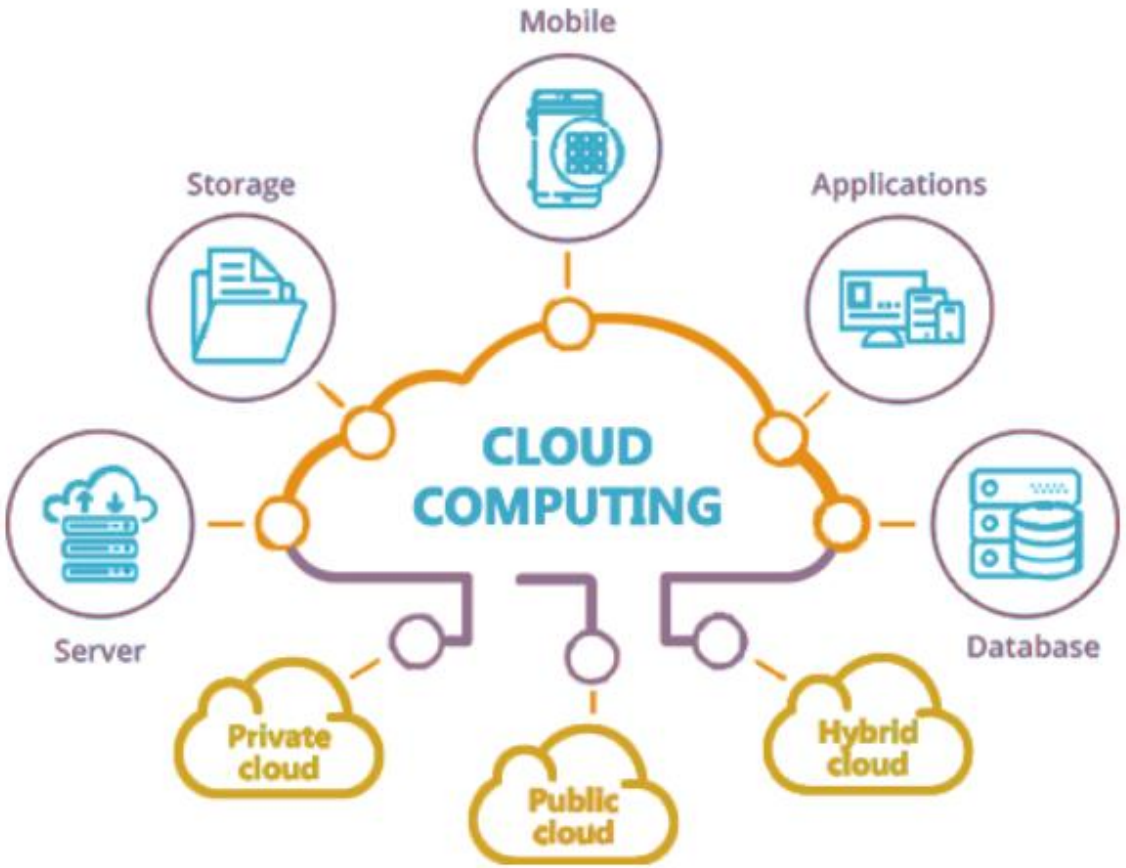
1. **Thinking in parallel**
   - “An introduction to parallel algorithms”, Joseph Jájá, Addison-Wesley (1992)
2. **Data stream processing**
   - “Algorithms for big data”, Moran Feldman, World Scientific (2020)
   - “Data stream algorithms”, lecture notes by Amit Chakrabarti

- Additional material that will be made available on Teams
  - Slides
  - Links to online free resources

Cloud Computing: an introduction

What is *Cloud Computing*?

The name *Cloud Computing* comes from the common use of a cloud-shaped symbol as an abstraction for the complex infrastructure it contains in system diagrams



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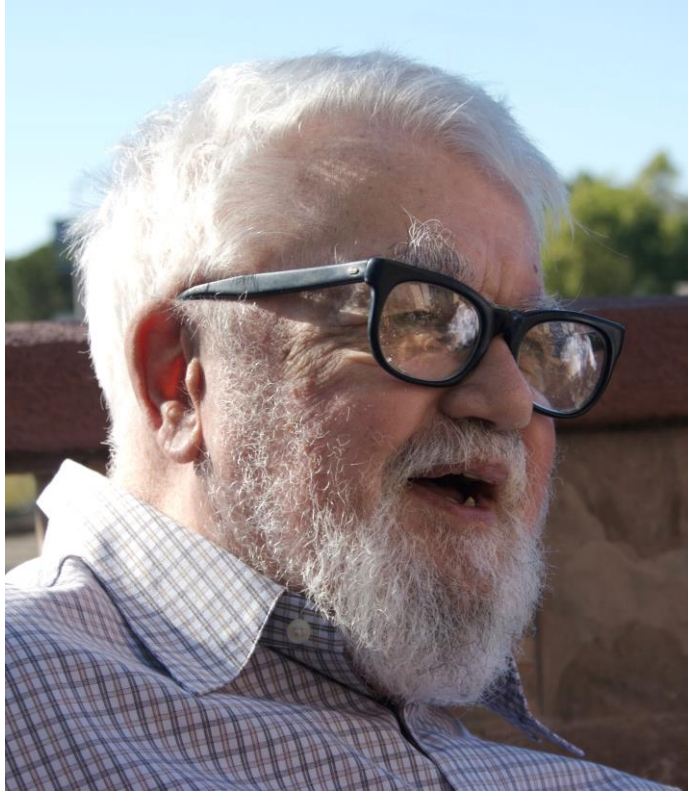

John McCarthy
(September 4, 1927 – October 24, 2011)

John McCarthy was both a computer and a cognitive scientist who provided fundamental contribution to the AI, and received the Turing Award in 1971

"If computers of the kind I have advocated become the computers of the future, then ***computing may*** *someday* ***be organized as a public utility*** *just as the telephone system is a public utility… The computer utility could become* ***the basis of a new and important industry.****"*

(John McCarthy, 1961)

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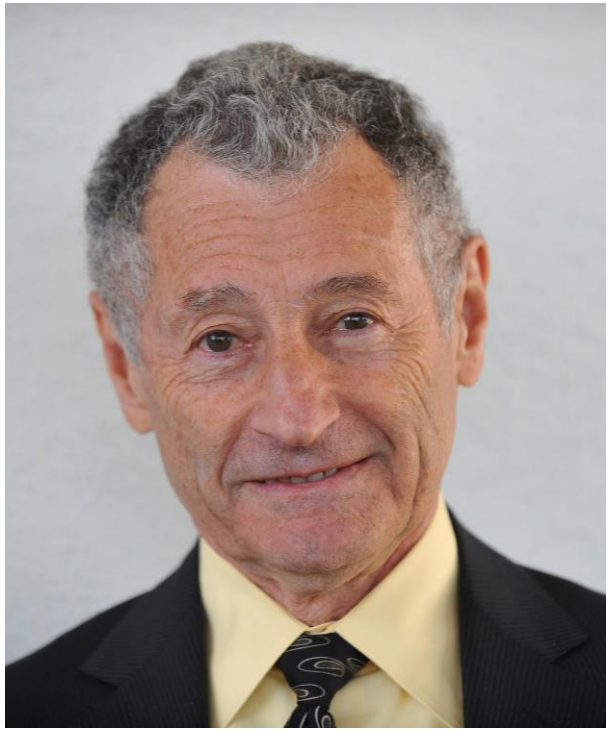

Leonard Kleinrock
(New York, June 13, 1934)

Leonard Kleinrock has been a chief scientist of the Advanced Research Projects Agency Network (ARPANET).

"as of now, computer networks are still in their infancy, but as they grow up and become sophisticated, ***we will*** *probably* ***see the spread of 'computer utilities' which****, like present electric and telephone utilities,* ***will service individual homes and offices across the country.****"*

(Leonard Kleinrock, 1969)

I need to grow my infrastructure, but I do not know for how long…
I have a lot of infrastructure that I want to rent …
I have a surplus of infrastructure that I want to make use of
I cannot invest in infrastructure, I just started my business….
I have infrastructure and middleware and I can host applications
Global Cloud Marketplace
I want to focus on application logic and not maintenance and scalability issues
I have infrastructure and provide application services
I want to access and edit my documents and photos from everywhere..

Examples of cloud computing commitment

- The *TimesMachine*
  - The New York Times had a huge number of scanned images of historical newspapers, ranging from 1851-1922
  - They want to process this set of images into separate articles in PDF format
  - They used 100 EC2 instances (Amazon Elastic Compute Cloud)
    - 24 hours computational work
    - $ 240 for computational power + $ 650 for data transfer and storage (Input: 4.0TB, Output: 1.5TB) = Total $ 890

  - The New York Times had the option to utilize 100 servers for 24 hours at the low standard cost of ten cent an hour for every server
  - If New York times had bought even a solitary server for this errand, the probable expense would have surpassed the $ 890 for simply the hardware

Examples of cloud computing commitment

- The Washington Post
  - were able to transform 17,481 pages of scanned document images into a searchable database
  - utilized 200 Amazon EC2 instances to conduct on the scanned files to create searchable text
  - The total cost of using 1,407 hours of virtual machine was $ 144.62
- The Defense Information Systems Agency (DISA)
  - has compared the cost of the usage of Amazon EC2 versus internally maintained servers
  - the price of growing a simple application known as the Tech Early Bird was $ 30,000 by using internally maintained servers
  - Amazon charged 10 cents an hour for the provider, and DISA paid a total of $5 to expand a software that matched the overall performance of the in-house application

What is *Cloud Computing*?

- use of computing resources (servers, storage, databases, software, analytics etc.)
  that are delivered as a service over a network (usually the Internet)
  to provide flexible resources, faster innovation and economies of scale
- the term *Cloud computing* refers to both the applications delivered as services (Software as a Service, SaaS) and *the cloud*, i.e., hardware and system software in the datacenters that provide those services (Hardware as a Service, HaaS)
- *The cloud* is a <u>parallel</u> and <u>distributed system</u>
  consisting of a collection of interconnected and virtualized computers
  that are presented as one or more unified computing resources
  based on service-level agreements established between the provider and consumers
  and that are mainly used to process <u>big</u> amount of <u>data</u>

Cloud computing and Big Data

- *Big data* and *cloud computing* are two distinctly different ideas …although they are almost inseparable
- *Cloud computing* is the processing of anything, including *big data* analytics, on the cloud. Cloud computing technologies analyze large data sets much more quickly than a standard computer
- *Big data*: huge sets of data that are usually far too large to be processed on a regular computer

Big data refers to things one can do at a large scale that cannot be done at a smaller one.

(Big Data: A Revolution - Mayer-Schonberger, Cukier)

- *Big data* refers to:

1. huge amount of data (structured, semi-structured, or unstructured) that is often derived from different sources
2. the act of processing enormous volumes of data that is analyzed through a set of mathematical algorithms via distributed computing software platforms (e.g., MapReduce, Hadoop, Spark, Databricks and Cloudera)

Pros and cons of big data in the cloud

PROS

- Scalability
  - public clouds manage hundreds of thousands of servers spread across a fleet of global data centers
- Agility
  - Users assemble the infrastructure of almost any size
- Cost
  - Users pay for what they use (pay-per-use model)
- Accessibility
  - Data and services are accessible from anywhere using the Internet
- Resilience
  - Clouds replicate data to maintain high availability in storage resources

CONS

- Network dependence
  - Network outages result in increasing latency at best, or complete cloud inaccessibility at worst
- Storage costs
  - Data storage in the cloud can present a substantial long-term cost for big data projects
- Security
  - Big data often involves sensitive data
- Lack of standardization
  - no single way to architect, implement or operate a big data deployment in the cloud

Some challenges in cloud computing

1. **Fast parallel processing of the data**
   - The lack of a standardized hardware and software architecture has brought to the development of several parallel models
   - PRAM provides a good basic theoretical framework
   - MPC is a more refined platform-agnostic model which, to some extent, emulates to MapReduce, Hadoop, and Spark, but its development is still at an early stage
2. **Storage cost optimization**
   - Compute useful information by storing only a small fraction of the input data and discard the rest (data streams)
3. Data security …and more